

## DIFFERENTIAL EQUATIONS AND THEIR CLASSIFICATIONS

**(9.1) Definition.** An equation involving one dependent variable and its derivatives with respect to one or more independent variables, is called a **differential equation**. For example

$$(i) \quad \frac{dy}{dx} + y \cos x = \sin x$$

$$(ii) \quad \frac{d^2y}{dx^2} + xy \left( \frac{dy}{dx} \right)^2 = 0$$

$$(iii) \quad \left[ 1 + \left( \frac{dy}{dx} \right)^2 \right]^{3/2} = \frac{d^2y}{dx^2}$$

$$(iv) \quad x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = nx$$

$$(v) \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

are differential equations.

**(9.2) Definition.** A differential equation, in which ordinary derivatives of the dependent variable with respect to a single independent variable occur, is called an **ordinary differential equation**. Equations (i), (ii) and (iii) above are examples of ordinary differential equations.

**(9.3) Definition.** A differential equation involving partial derivatives of the dependent variable with respect to more than one independent variable is called a **partial differential equation**. Equations (iv) and (v) given above are partial differential equations.

**(9.4) Definition.** The **order** of a differential equation is the order of the highest derivative that occurs in the equation.

**(9.5) Definition.** The **degree** of a differential equation is the degree of the highest order derivative that appears in the equation. ( The dependent variable and its derivatives should be expressed in a form free from radicals and fractions).

The differential equations given in (9.1) have the following orders and degrees :

- (i) order 1, degree 1
- (ii) order 2, degree 1
- (iii) order 2, degree 2
- (iv) order 1, degree 1
- (v) order 2, degree 1.

**Example 2.** Form the differential equation by eliminating two constants  $A$  and  $B$  from the relation

$$y = A \sin x + B \cos x \quad (1)$$

**Solution.** It is clear that three equations are required to eliminate two unknowns  $A$  and  $B$ . We obtain two other needed equations by successive differentiation of (1). Thus

$$\frac{dy}{dx} = A \cos x - B \sin x \quad (2)$$

$$\frac{d^2y}{dx^2} = -A \sin x - B \cos x \quad (3)$$

From (1), (2) and (3) we obtain

$$\frac{d^2y}{dx^2} + y = 0 \quad (4)$$

which is the required differential equation. (1) is the general solution of (4).

**Example 3.** Find the differential equation of all parabolas whose axes are parallel to the axis of  $y$ .

**Solution.** General equation of a parabola whose axis is parallel to the  $y$ -axis is

$$y = ax^2 + bx + c \quad (1)$$

In order to obtain the differential equation, we have to eliminate  $a$ ,  $b$ ,  $c$  from (1) and three more equations. Differentiating (1) successively, we have

$$\frac{dy}{dx} = 2ax + b,$$

$$\frac{d^2y}{dx^2} = 2a,$$

$$\frac{d^3y}{dx^3} = 0$$

The last equation does not contain any of the constants  $a$ ,  $b$  and  $c$ . Thus

$$\frac{d^3y}{dx^3} = 0$$

is the differential equation of all parabolas whose axes are parallel to the axis of  $y$ .

Classify each of the following equations as ordinary or partial differential equations : state the order and the degree of each equation and determine whether the equation is linear or nonlinear :

$$(i) \quad \frac{d^3 y}{dx^3} + 4 \frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 3y = \cos x$$

$$(ii) \quad x^2 dy + y^2 dx = 0$$

$$(iii) \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

$$(iv) \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + u \frac{\partial u}{\partial x} + u \frac{\partial u}{\partial y} + u = 0$$

$$(v) \quad \left( \frac{dy}{dx} \right)^2 = \left( \frac{d^2 y}{dx^2} + y \right)^{\frac{3}{2}}$$

Determine whether each of the following differential equations is or is not separable. Note: An equation of the form  $F(x)G(y)dx + f(x)g(y)dy = 0 \dots (*)$  is called an equation with variables separable or simply a separable equation. Equation (\*) can be restated  $\frac{F(x)}{f(x)}dx + \frac{g(y)}{G(y)}dy = 0$ .

1.  $\frac{dy}{dx} = 3y^2 - y^2 \sin(x)$

**Solution:**

Since

$$\frac{dy}{dx} = 3y^2 - y^2 \sin(x)$$

$$\Leftrightarrow \frac{dy}{dx} = y^2(3 - \sin(x))$$

$$\Leftrightarrow \frac{1}{y^2} dy = (3 - \sin(x))dx$$

$$\Leftrightarrow (3 - \sin(x))dx - \frac{1}{y^2} dy = 0,$$

we can conclude that the differential equation  $\frac{dy}{dx} = 3y^2 - y^2 \sin(x)$  is separable.

2.  $\frac{dy}{dx} = 3x - y \sin(x)$

**Solution:**

Since

$$\frac{dy}{dx} = 3x - y \sin(x)$$

$$\Leftrightarrow dy = (3x - y \sin(x))dx$$

$$\Leftrightarrow (y \sin(x) - 3x)dx - dy = 0,$$

we can conclude that the differential equation  $\frac{dy}{dx} = 3x - y \sin(x)$  is not separable.

3.  $x \frac{dy}{dx} = (x - y)^2$

**Solution:**

Since

$$x \frac{dy}{dx} = (x - y)^2$$

$$\Leftrightarrow x dy = (x - y)^2 dx$$

$$\Leftrightarrow \frac{1}{(x - y)^2} dy = \frac{1}{x} dx$$

$$\Leftrightarrow \frac{1}{(x - y)^2} dy - \frac{1}{x} dx = 0,$$

we can say that the differential equation  $x \frac{dy}{dx} = (x - y)^2$  is not separable.

4.  $\frac{dy}{dx} = \sqrt{1 + x^2}$

**Solution:**

Since

$$\frac{dy}{dx} = \sqrt{1+x^2}$$

$$\Leftrightarrow dy - \sqrt{1+x^2} dx = 0,$$

we can conclude that the differential equation  $\frac{dy}{dx} = \sqrt{1+x^2}$  is separable.

5.  $\frac{dy}{dx} + 4y = 8$

**Solution:**

Since

$$\frac{dy}{dx} + 4y = 8$$

$$\Leftrightarrow \frac{dy}{dx} = 8 - 4y$$

$$\Leftrightarrow dy = (8 - 4y)dx$$

$$\Leftrightarrow \frac{1}{8-4y} dy - dx = 0$$

we can conclude that the differential equation  $\frac{dy}{dx} + 4y = 8$  is separable.

6.  $\frac{dy}{dx} + xy = 4x$

**Solution:**

$$\frac{dy}{dx} + xy = 4x$$

$$\Leftrightarrow \frac{dy}{dx} = 4x - xy$$

$$\Leftrightarrow \frac{dy}{dx} = x(4 - y)$$

$$\Leftrightarrow \frac{1}{4-y} dy = x dx$$

$$\Leftrightarrow \frac{1}{4-y} dy - x dx = 0,$$

we can conclude that the differential equation  $\frac{dy}{dx} + xy = 4x$  is separable.

7.  $\frac{dy}{dx} + 4y = x^2$

**Solution:**

Since

$$\frac{dy}{dx} + 4y = x^2$$

$$\Leftrightarrow \frac{dy}{dx} = x^2 - 4y$$

$$\Leftrightarrow dy = (x^2 - 4y)dx$$

$$\Leftrightarrow dy - (x^2 - 4y)dx = 0,$$

we can say that the differential equation  $\frac{dy}{dx} + 4y = x^2$  is not separable.

8.  $\frac{dy}{dx} = xy - 3x - 2y + 6$

**Solution:**

Since

$$\frac{dy}{dx} = xy - 3x - 2y + 6$$

$$\Leftrightarrow \frac{dy}{dx} = (x - 2)(y - 3)$$

$$\Leftrightarrow \frac{1}{y - 3} dy - (x - 2)dx = 0,$$

we can say that the differential equation  $\frac{dy}{dx} = xy - 3x - 2y + 6$  is separable.

9.  $\frac{dy}{dx} = \sin(x + y)$

**Solution:**

Since

$$\frac{dy}{dx} = \sin(x + y)$$

$$\Leftrightarrow dy - \sin(x + y) dx = 0,$$

we can say that the differential equation  $\frac{dy}{dx} = \sin(x + y)$  is not separable.

10.  $y \frac{dy}{dx} = e^{x-3y^2}$

**Solution:**

Since

$$y \frac{dy}{dx} = e^{x-3y^2}$$

$$\Leftrightarrow y \frac{dy}{dx} = \frac{e^x}{e^{3y^2}}$$

$$\Leftrightarrow ye^{3y^2} dy - e^x dx = 0,$$

we can conclude that the differential equation  $y \frac{dy}{dx} = e^{x-3y^2}$  is separable.

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## SEPARABLE EQUATIONS

**(9.11) Definition.** A differential equation of the type

$$F(x) G(y) dx + f(x) g(y) dy = 0 \quad (1)$$

is called an equation with **variables separable** or simply a **separable equation**. Equation (1) may be written as

$$\frac{F(x)}{f(x)} dx + \frac{g(y)}{G(y)} dy = 0$$

which can be easily integrated.

**Example 8. Solve:**  $\frac{dy}{dx} = \frac{x^2}{y}$  (1)

**Solution.** Equation (1) can be written as

$$y dy = x^2 dx$$

Integration gives

$$\frac{y^2}{2} = \frac{x^3}{3} + c_1$$

or,  $3y^2 = 2x^3 + c,$

which is the required solution of (1).

**Example 9.** Solve:  $\frac{dy}{dx} = \frac{1}{x \tan y}$

**Solution.** The given equation is separable and can be written as

$$\frac{dx}{x} = \tan y \, dy$$

Integrating, we have

$$\ln |x| = -\ln |\cos y| + c$$

or,  $\ln |x| + \ln |\cos y| = c$

$$\ln |x \cos y| = c$$

or,  $|x \cos y| = e^c = a \text{ (say)}$

is the required solution.

**Example 10.** Solve:  $x \sin y \, dx + (x^2 + 1) \cos y \, dy = 0$ . (1)

**Solution.** Dividing (1) by  $(x^2 + 1) \sin y$ , we have

$$\frac{x}{x^2 + 1} \, dx + \cot y \, dy = 0$$

which is a separable equation. Therefore,

$$\int \frac{x \, dx}{x^2 + 1} + \int \cot y \, dy = c_1 = \ln |c_2|$$

i.e.,  $\frac{1}{2} \ln(x^2 + 1) + \ln |\sin y| = \ln |c_2|$

**Example 11.** Solve the initial value problem

$$\frac{dy}{dx} = \frac{2x}{y + x^2 y}, \quad y(0) = -2$$

**Solution.** We have

$$\frac{dy}{dx} = \frac{2x}{y(1 + x^2)}$$

or  $y \, dy = \frac{2x}{1 + x^2} \, dx$

On integrating, we obtain

$$\frac{y^2}{2} = \ln(1 + x^2) + c$$

$$y = \pm [2 \ln(1 + x^2) + c]^{1/2}$$

Now,  $y(0) = -2$

$$= \pm [2 \ln 1 + c]^{1/2}$$

Since  $y(0) = -2$ ,

we must take negative sign of the radical and this gives  $c = 4$ .

Hence

$$y = - [2 \ln(1 + x^2) + 4]^{1/2}$$

is the required solution.

## SEPARABLE DIFFERENTIAL EQUATION

Q#01:  $\frac{dy}{dx} = \frac{x^2}{y(1+x^3)}$

Sol: Step#01: Separate the variables

$$y dy = \frac{x^2}{1+x^3} dx$$

Step#02: integrating both sides

$$\int y dy = \int \frac{x^2}{1+x^3} dx \quad \text{--- (A)}$$

$$\text{let } t = 1+x^3$$

$$dt = 3x^2 dx$$

$$\left[ \frac{dt}{3} = x^2 dx \right]$$

putting the values in eq (A)

$$\int y dy = \int \frac{1}{t} \frac{dt}{3}$$

$$\frac{y^2}{2} = \frac{1}{3} \ln(t) + C$$

$$\therefore \left[ t = 1+x^3 \right]$$

So above Equation becomes.

$$\left[ \frac{y^2}{2} = \frac{1}{3} \ln(1+x^3) + C \right]$$

Q#02:  $(xy+2x+y+2)dx + (x^2+2x)dy = 0$

Sol: Step#01: Separate the variables

$$[x(y+2) + 1(y+2)]dx + (x^2+2x)dy = 0$$

$$(y+2)(x+1)dx + x(x+2)dy = 0$$

$$(y+2)(x+1)dx = -x(x+2)dy$$

$$\frac{(x+1)dx}{x(x+2)} = \frac{-1}{(y+2)} dy$$

Step#02: Integrating both sides.

$$\int \frac{(x+1)}{x(x+2)} dx = - \int \frac{1}{(y+2)} dy \quad \text{--- (A)}$$

Consider:  $\frac{x+1}{x(x+2)} = \frac{A}{x} + \frac{B}{x+2} \quad \text{--- (B)}$

$$\frac{x+1}{x(x+2)} = \frac{(x+2)A + xB}{x(x+2)}$$

$$\boxed{x+1 = (x+2)A + xB} \quad \text{--- (1)}$$

For A=? put  $x=0$  in eq (1)

$$0+1 = (0+2)A + 0$$

$$\boxed{A = \frac{1}{2}}$$

For B=? put  $x = -2$  in eq-①

$$-2+1 = (-2+2)A + (-2)B$$

$$-1 = 0 - 2B$$

$$\boxed{B = \frac{1}{2}}$$

Putting the values of A and B in eq-②

$$\frac{x+1}{x(x+2)} = \frac{1/2}{x} + \frac{1/2}{x+2}$$

$$\boxed{\frac{x+1}{x(x+2)} = \frac{1}{2} \cdot \frac{1}{x} + \frac{1}{2} \cdot \frac{1}{x+2}}$$

put above values in eq-③

$$\frac{1}{2} \int \frac{1}{x} dx + \frac{1}{2} \int \frac{1}{x+2} dx = - \int \frac{1}{y+2} dy$$

$$\frac{1}{2} \ln(x) + \frac{1}{2} \ln(x+2) = - \ln(y+2) + C$$

$$\therefore \boxed{\ln A^n = n \ln A}$$

$$\ln x^{1/2} + \ln (x+2)^{1/2} = \ln (y+2)^{-1} + C$$

$$\boxed{\ln \sqrt{x} + \ln \sqrt{x+2} = \ln \left( \frac{1}{y+2} \right) + C}$$

## SEPARABLE DIFFERENTIAL EQ

(3)  $\frac{dy}{dx} + y^2 \sin x = 0$

Sol: Step#01: Separate the variables

$$\frac{dy}{dx} = -y^2 \sin x$$

$$\frac{dy}{y^2} = -\sin x \, dx$$

Step#02: Integrating both sides

$$\int \frac{dy}{y^2} = -\int \sin x \, dx$$

$$\frac{y^{-2+1}}{-1} = +\cos x + C$$

$$\boxed{-\frac{1}{y} = \cos x + C}$$

(4)  $\frac{dy}{dx} = 1 + x + y^2 + xy^2$

Sol: Step#01: Separate the variables

$$\frac{dy}{dx} = 1(1+x) + y^2(1+x)$$

$$\frac{dy}{dx} = (1+x)(1+y^2)$$

$$\frac{dy}{1+y^2} = (1+x) \, dx$$

Step#02: Integrating both sides

$$\int \frac{dy}{1+y^2} = \int (1+x) dx$$

$$\therefore \boxed{\tan^{-1}(y) = \frac{1}{1+y^2}}$$

So,

$$\boxed{\tan^{-1}(y) = x + \frac{x^2}{2} + C}$$

⑤  $\frac{dy}{dx} = 2x^2 + y - x^2y + xy - 2x - 2$

Sol.

Step#01: Separate the variables

$$\frac{dy}{dx} = x^2(2-y) + (y-2) + x(y-2)$$

$$\frac{dy}{dx} = -x^2(y-2) + (y-2) + x(y-2)$$

$$\frac{dy}{dx} = (y-2)(-x^2 + 1 + x)$$

$$\frac{dy}{(y-2)} = \frac{dx(-x^2 + 1 + x)}{1}$$

Step#02: Integrating both sides

$$\int \frac{dy}{(y-2)} = \int (-x^2 + 1 + x) dx$$

$$\ln(y-2) = -\frac{x^3}{3} + x + \frac{x^2}{2} + C$$

⑥  $\operatorname{cosec} y \, dx + \sec x \, dy = 0$

Sol: Step#01: Separate the variables

$$\operatorname{cosec} y \, dx = -\sec x \, dy$$

$$\frac{dx}{\sec x} = \frac{-dy}{\operatorname{cosec} y}$$

Step#02: Integrating both sides

$$\int \frac{dx}{\sec x} = -\int \frac{dy}{\operatorname{cosec} y}$$

$$\int \cos x \, dx = -\int \sin y \, dy$$

$$\sin x = -(-\cos y) + C$$

$$\sin x = \cos y + C$$

⑦  $y(1+x) \, dx + x(1+y) \, dy = 0$

Sol: Step#01: Separate the variables

$$y(1+x) \, dx = -x(1+y) \, dy$$

$$\frac{(1+x)}{x} dx = -\frac{(1+y)}{y} dy$$

Step#02: Integrating both sides

$$\int \frac{(1+x)}{x} dx = -\int \frac{(1+y)}{y} dy$$

$$\int \frac{1}{x} dx + \int 1 dx = -\int \frac{1}{y} dy - \int 1 dy$$

$$\ln x + x = -\ln y - y + C$$

$$(8) \quad \frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$$

Sol. Step#01: Separate the variables

$$\frac{dy}{dx} = -\sqrt{\frac{1-y^2}{1-x^2}}$$

$$\frac{dy}{dx} = \frac{-\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

$$\frac{dy}{\sqrt{1-y^2}} = \frac{-dx}{\sqrt{1-x^2}}$$

Step#02: Integrating both sides

$$\int \frac{dy}{\sqrt{1-y^2}} = - \int \frac{dx}{\sqrt{1-x^2}}$$

$$\therefore \boxed{\sin^{-1} y = \frac{1}{\sqrt{1-x^2}}}$$

$$\text{So, } \boxed{\sin^{-1} y = -\sin^{-1} x + C}$$

$$\textcircled{9} (\sin x + \cos x) dy + (\cos x - \sin x) dx = 0$$

Sol: Step #01: Separate the variables

$$(\sin x + \cos x) dy = -(\cos x - \sin x) dx$$

$$dy = \frac{-(\cos x - \sin x)}{(\sin x + \cos x)} dx$$

Step #02: Integrating both sides

$$\int dy = - \int \frac{(\cos x - \sin x) dx}{(\sin x + \cos x)} \quad \text{--- (A)}$$

$$\text{Let } t = \sin x + \cos x$$

$$dt = (\cos x - \sin x) dx$$

putting the values in eq (A)

$$\int dy = - \int \frac{1}{t} dt$$

$$y = -\ln(t) + C$$

$$\therefore \boxed{t = \sin u + \cos u}$$

$$\boxed{y = -\ln(\sin u + \cos u) + C}$$

(10)  $(2x \cos y) dx + x^2 (\sec y - \sin y) dy = 0$

Sol: Step #01: Separate the variables

$$(2x \cos y) dx = -x^2 (\sec y - \sin y) dy$$

$$\left(\frac{2x}{x^2}\right) dx = -\left(\frac{\sec y - \sin y}{\cos y}\right) dy$$

Step #02: Integrating both sides

$$\int \frac{2}{x} dx = \int \left( \frac{\sin y}{\cos y} - \frac{\sec y}{\cos y} \right) dy$$

$$2 \ln x = \int \tan y dy - \int \frac{1}{\cos y} dy$$

$$2 \ln x = \ln \sec y - \int \frac{1}{\cos^2 y} dy$$

$$2 \ln x = \ln \sec y - \int \sec^2 y dy$$

$$\boxed{2 \ln x = \ln \sec y - \tan y + C}$$

Q# Solve the initial value problems.

①  $(3x+8)(y^2+4)dx - 4y(x^2+5x+6)dy=0$   
 $y(1)=2$

Sol. Step#01: Separate the variables

$$(3x+8)(y^2+4)dx = 4y(x^2+5x+6)dy$$

$$\frac{(3x+8)}{(x^2+5x+6)} dx = \frac{4y}{y^2+4} dy$$

Step#02: Integrating both sides

$$\int \frac{(3x+8)}{(x^2+5x+6)} dx = \int \frac{4y}{y^2+4} dy$$

$$\int \frac{(3x+8)}{x^2+2x+3x+6} dx = \int \frac{4y}{y^2+4} dy$$

$$\int \frac{(3x+8)}{x(x+2)+3(x+2)} dx = \int \frac{4y}{y^2+4} dy$$

$$\boxed{\int \frac{(3x+8)}{(x+2)(x+3)} dx = \int \frac{4y}{y^2+4} dy} \quad \text{--- (A)}$$

(h)

Consider:  $\frac{3x+8}{(x+2)(x+3)} = \frac{A}{x+2} + \frac{B}{x+3}$  — (1)

$$\frac{3x+8}{(x+2)(x+3)} = \frac{(x+3)A + (x+2)B}{(x+2)(x+3)}$$

$$3x+8 = (x+3)A + (x+2)B$$

for  $B=?$  put  $x = -3$

$$3(-3)+8 = (-3+3)A + (-3+2)B$$

$$-9+8 = 0 - B \Rightarrow \boxed{B=1}$$

for  $A=?$  put  $x = -2$

$$3(-2)+8 = (-2+3)A + (-2+2)B$$

$$2 = A + 0 \Rightarrow \boxed{A=2}$$

putting the values in eq (1)

$$\frac{3x+8}{(x+2)(x+3)} = \frac{2}{x+2} + \frac{1}{x+3}$$

put this value in eq (A)

$$\int \frac{2}{x+2} dx + \int \frac{1}{x+3} dx = 2 \int \frac{2y}{y^2+4} dy$$

$$2 \ln(x+2) + \ln(x+3) = 2 \ln(y^2+4) + C$$

$$\boxed{2 \ln(x+2) + \ln(x+3) = 2 \ln(y^2+4) + C}$$

$$2. \frac{dy}{dx} = 3y^2 - y^2 \sin(x)$$

**Solution:**

Since  $\frac{dy}{dx} = y^2(3 - \sin(x)) \Leftrightarrow \frac{1}{y^2} dy - (3 - \sin(x))dx = 0$ , we obtain that

$$\int \frac{1}{y^2} dy - \int (3 - \sin(x)) dx = C_1$$

$$\Leftrightarrow -\frac{1}{y} - (3x + \cos(x)) = C_1$$

$$\Leftrightarrow -\frac{1}{y} = 3x + \cos(x) + C_1$$

$$\Leftrightarrow y = -\frac{1}{3x + \cos(x) + C_1}, \text{ let } C = -C_1$$

$$\Leftrightarrow y = \frac{1}{C - 3x - \cos(x)}$$

as the general solution for the differential equation  $\frac{dy}{dx} = 3y^2 - y^2 \sin(x)$ .

$$3. \frac{dy}{dx} = xy - 3x - 2y + 6$$

**Solution:**

Since  $\frac{dy}{dx} = xy - 3x - 2y + 6 \Leftrightarrow \frac{dy}{dx} = (x - 2)(y - 3) \Leftrightarrow \frac{1}{y-3} dy - (x - 2)dx = 0$ ,

we obtain

$$\int \frac{1}{y-3} dy - \int (x - 2) dx = C_1$$

$$\Leftrightarrow \ln|y - 3| - \frac{1}{2}x^2 + 2x = C_1$$

$$\Leftrightarrow \ln|y - 3| = \frac{1}{2}x^2 - 2x + C_1$$

$$\Leftrightarrow |y - 3| = e^{\frac{1}{2}x^2 - 2x + C_1}, \text{ let } C = e^{C_1}$$

$$\Leftrightarrow |y - 3| = Ce^{\frac{1}{2}x^2 - 2x}$$

as the general solution for differential equation  $\frac{dy}{dx} = xy - 3x - 2y + 6$ .

Note:

$$\text{Since } |y - 3| = Ce^{\frac{1}{2}x^2 - 2x},$$

4.  $\frac{dy}{dx} = \tan(y)$

**Solution:**

Since  $\frac{dy}{dx} = \tan(y) \Leftrightarrow \frac{1}{\tan(y)} dy - dx = 0$ , assume  $\tan(y) \neq 0$ , we obtain

$$\int \frac{1}{\tan(y)} dy - \int dx = C_1$$

$$\Leftrightarrow \int \frac{\cos(y)}{\sin(y)} dy - \int dx = C_1$$

$$\Leftrightarrow \int \frac{1}{\sin(y)} d(\sin(y)) - \int dx = C_1$$

$$\Leftrightarrow \ln |\sin(y)| - x = C_1$$

$$\Leftrightarrow \ln |\sin(y)| = x + C_1$$

$$\Leftrightarrow |\sin(y)| = e^{x+C_1}, \text{ let } C = e^{C_1}$$

$$\Leftrightarrow |\sin(y)| = Ce^x$$

as the general solution for  $\frac{dy}{dx} = \tan(y)$ .

Note:

Since  $|\sin(y)| = Ce^x$ ,

$$\sin(y) = \begin{cases} Ce^x & \text{if } 0 \leq \sin(y) \leq 1 \\ -Ce^x = Ke^x & \text{if } -1 \leq \sin(y) < 0 \end{cases} \Leftrightarrow y = \begin{cases} \arcsin(Ce^x) \\ \arcsin(Ke^x) \end{cases}$$

5.  $\frac{dy}{dx} = \frac{y}{x}$

**Solution:**

Since  $\frac{dy}{dx} = \frac{y}{x} \Leftrightarrow \frac{1}{y} dy - \frac{1}{x} dx = 0$ , we obtain

$$\int \frac{1}{y} dy - \int \frac{1}{x} dx = C_1$$

$$\Leftrightarrow \ln|y| - \ln|x| = C_1$$

$$\Leftrightarrow \ln|y| = \ln|x| + \ln e^{C_1}$$

$$\Leftrightarrow \ln|y| = \ln(e^{C_1}|x|), \text{ let } C = e^{C_1}$$

$$\Leftrightarrow |y| = C|x|$$

as the general solution for the differential equation  $\frac{dy}{dx} = \frac{y}{x}$ .

6.  $\frac{dy}{dx} = \frac{6x^2+4}{3y^2-4y}$

**Solution:**

Since  $\frac{dy}{dx} = \frac{6x^2+4}{3y^2-4y} \Leftrightarrow (3y^2 - 4y)dy - (6x^2 + 4)dx = 0$ , assume  $3y^2 - 4y \neq 0$  we obtain

$$\int (3y^2 - 4y) dy - \int (6x^2 + 4) dx = C$$

$$\Leftrightarrow y^3 - 2y^2 - 2x^3 - 4x = C$$

$$\Leftrightarrow y = \frac{2x^3 + 4x + C}{y^2 - 2y}$$

as the general solution for the differential equation  $\frac{dy}{dx} = \frac{6x^2+4}{3y^2-4y}$ .

7.  $(x^2 + 1) \frac{dy}{dx} = y^2 + 1$

**Solution:**

Since  $(x^2 + 1) \frac{dy}{dx} = y^2 + 1 \Leftrightarrow \frac{1}{1+y^2} dy - \frac{1}{1+x^2} dx = 0$ , we obtain

$$\int \frac{1}{1+y^2} dy - \int \frac{1}{1+x^2} dx = C$$

$$\Leftrightarrow \arctan(y) - \arctan(x) = C$$

$$\Leftrightarrow \arctan(y) = \arctan(x) + C$$

$$\Leftrightarrow y = \tan(\arctan(x) + C)$$

as the general solution for differential equation  $(x^2 + 1) \frac{dy}{dx} = y^2 + 1$ .

8.  $(y^2 - 1) \frac{dy}{dx} = 4xy^2$

**Solution:**

Since  $(y^2 - 1) \frac{dy}{dx} = 4xy^2 \Leftrightarrow \frac{y^2-1}{y^2} dy - 4x dx = 0$ , assume  $y \neq 0$ , we obtain

$$\int \left( \frac{y^2-1}{y^2} \right) dy - \int 4x dx = C$$

$$\Leftrightarrow \int \left( 1 - \frac{1}{y^2} \right) dy - \int 4x dx = C$$

$$\Leftrightarrow y + \frac{1}{y} - 2x^2 = C$$

$$\Leftrightarrow \frac{y^2 + 1}{y} = 2x^2 + C$$

as the general solution for the differential equation  $(y^2 - 1) \frac{dy}{dx} = 4xy^2$ .

9.  $\frac{dy}{dx} = e^{-y}$

**Solution:**

Since  $\frac{dy}{dx} = e^{-y} \Leftrightarrow \frac{1}{e^{-y}} dy - dx = 0 \Leftrightarrow e^y dy - dx = 0$ , we obtain

$$\int e^y dy - \int dx = C$$

$$\Leftrightarrow e^y - x = C$$

$$\Leftrightarrow e^y = x + C$$

$$\Leftrightarrow y = \ln|x + C|$$

as the general solution for differential equation  $\frac{dy}{dx} = e^{-y}$ .

10.  $\frac{dy}{dx} = e^{-y} + 1$

**Solution:**

Since  $\frac{dy}{dx} = e^{-y} + 1 \Leftrightarrow \frac{1}{e^{-y}+1} dy - dx = 0$ , we obtain

$$\int \frac{1}{e^{-y}+1} dy - \int dx = C_1$$

$$\Leftrightarrow \int \left( \frac{e^y}{e^y + 1} \right) dy - \int dx = C_1$$

$$\Leftrightarrow \int \frac{1}{e^y + 1} d(e^y) - \int dx = C_1$$

$$\Leftrightarrow \ln(e^y + 1) - x = C_1$$

$$\Leftrightarrow \ln(e^y + 1) = x + C_1$$

$$\Leftrightarrow e^y = e^{x+C_1} - 1, \text{ let } C = e^{C_1}$$

$$\Leftrightarrow y = \ln |Ce^x - 1|$$

as the general solution for differential equation  $\frac{dy}{dx} = e^{-y} + 1$ .

11.  $\frac{dy}{dx} = 3xy^3$

**Solution:**

Since  $\frac{dy}{dx} = 3xy^3 \Leftrightarrow \frac{1}{y^3} dy - 3x dx = 0$ , assume  $y \neq 0$ , we obtain

$$\int \frac{1}{y^3} dy - \int 3x dx = C_1$$

$$\Leftrightarrow \frac{1}{-2y^2} - \frac{3}{2}x^2 = C_1$$

$$\Leftrightarrow \frac{1}{y^2} = -3x^2 - 2C_1, \text{ let } C = -2C_1$$

$$\Leftrightarrow y^2 = \frac{1}{C - 3x^2}$$

As the general solution for differential equation  $\frac{dy}{dx} = 3xy^3$ .

12.  $\frac{dy}{dx} = \frac{2+\sqrt{x}}{2+\sqrt{y}}$

**Solution:**

Since  $\frac{dy}{dx} = \frac{2+\sqrt{x}}{2+\sqrt{y}} \Leftrightarrow (2 + \sqrt{y})dy - (2 + \sqrt{x})dx = 0$ , we obtain

$$\int (2 + \sqrt{y})dy - \int (2 + \sqrt{x})dx = C_2$$

$$\Leftrightarrow 2y + \frac{2}{3}y\sqrt{y} - 2x - \frac{2}{3}x\sqrt{x} = C_2$$

$$\Leftrightarrow 2y \left( 1 + \frac{1}{3}\sqrt{y} \right) = 2x \left( 1 + \frac{1}{3}\sqrt{x} \right) + C_2$$

$$\Leftrightarrow y \left( \frac{3 + \sqrt{y}}{3} \right) = x \left( \frac{3 + \sqrt{x}}{3} \right) + \frac{C_2}{2}, \text{ let } C_1 = \frac{C_2}{2}$$

$$\Leftrightarrow y = \frac{x \left( \frac{3 + \sqrt{x}}{3} \right) + C_1}{\frac{3 + \sqrt{y}}{3}}, \text{ let } C = 3C_1$$

$$\Leftrightarrow y = \frac{x(3 + \sqrt{x}) + C}{3 + \sqrt{y}}$$

as the general solution for the differential equation  $\frac{dy}{dx} = \frac{2+\sqrt{x}}{2+\sqrt{y}}$ .

$$13. \frac{dy}{dx} - 3x^2y^2 = -3x^2$$

**Solution:**

Since  $\frac{dy}{dx} - 3x^2y^2 = -3x^2 \Leftrightarrow \frac{dy}{dx} - 3x^2(y^2 - 1) = 0 \Leftrightarrow \frac{1}{y^2-1} dy - 3x^2 dx = 0$ , we obtain

$$\int \frac{1}{y^2-1} dy - \int 3x^2 dx = C$$

$$\Leftrightarrow \int \left( \frac{\frac{1}{2}}{y-1} - \frac{\frac{1}{2}}{y+1} \right) dy - \int 3x^2 dx = C_1$$

$$\Leftrightarrow \frac{1}{2} \ln|y-1| - \frac{1}{2} \ln|y+1| - x^3 = C_1$$

$$\Leftrightarrow \ln \left| \frac{y-1}{y+1} \right| = 2(x^3 + C_1)$$

$$\Leftrightarrow \left| \frac{y-1}{y+1} \right| = e^{2x^3+C}, \text{ let } C = 2C_1$$

$$\Leftrightarrow |y-1| = |y+1|e^{2x^3+C}$$

as the general solution from the differential equation  $\frac{dy}{dx} - 3x^2y^2 = -3x^2$ .

$$14. \frac{dy}{dx} - 3x^2y^2 = 3x^2$$

**Solution:**

Since  $\frac{dy}{dx} - 3x^2y^2 = 3x^2 \Leftrightarrow \frac{dy}{dx} = 3x^2(1 + y^2) \Leftrightarrow \frac{1}{1+y^2} dy - 3x^2 dx = 0$ , we obtain

$$\int \frac{1}{1+y^2} dy - \int 3x^2 dx = C$$

$$\Leftrightarrow \arctan y - x^3 = C$$

$$\Leftrightarrow y = \tan(x^3 + C)$$

as the general solution of differential equation  $\frac{dy}{dx} - 3x^2y^2 = 3x^2$ .

$$15. \frac{dy}{dx} = 200y - 2y^2$$

**Solution:**

Since  $\frac{dy}{dx} = 200y - 2y^2 \Leftrightarrow \frac{1}{200y-2y^2} dy = dx \Leftrightarrow \frac{1}{200y-2y^2} dy - dx = 0$ , assume

$y \neq 100$ , we obtain

$$\int \frac{1}{200y-2y^2} dy - \int dx = C_2$$

$$\Leftrightarrow \int \frac{1}{2y(100-y)} dy - \int dx = C_2$$

$$\Leftrightarrow \int \left( \frac{\frac{1}{100}}{2y} + \frac{\frac{1}{200}}{(100-y)} \right) dy - \int dx = C_2$$

$$\Leftrightarrow \frac{1}{200} \ln|y| - \frac{1}{200} \ln|100-y| - x = C_2$$

$$\Leftrightarrow \frac{1}{200} \ln \left| \frac{y}{100-y} \right| = x + C_2$$

Solve the initial value problems :

16.  $2(y - 1) dy = (3x^2 + 4x + 2) dx, y(0) = -1$
17.  $(3x + 8)(y^2 + 4) dx - 4y(x^2 + 5x + 6) dy = 0, y(1) = 2$
18.  $(1 + 2y^2) dy = y \cos x dx, y(0) = 1$
19.  $8 \cos^2 y dx + \csc^2 x dy = 0, y\left(\frac{\pi}{12}\right) = \frac{\pi}{4}$
20.  $\frac{dy}{dx} = \frac{x(x^2 + 1)}{4y^2}, y(0) = -\frac{1}{\sqrt{2}}$